

SCIENTIFIC METHOD

We employ the scientific method throughout this chapter to explore phenomena and observations, and pursue truth and the quest for greater knowledge. The steps are:

1. Ask a Question
2. Research the Problem
3. Construct a Hypothesis
4. Test the Hypothesis
5. Analyze the Data
6. Conclusions and Communication

Step 1: Ask a Question

A second grader may ask “Will my plant grow better using a white light bulb or a yellow light bulb?” A Ph.D. candidate may ask “What is the best monetary policy to stimulate GDP and hold back inflation in a new electronic economy with no country borders and minimal barriers to entry?” A Wall Street quant may simply ask “How much will my order move the market?”

Step 2: Research the Problem

The second step is to learn as much as possible about the problem. It is important to identify what worked and what failed in order to avoid reinventing the wheel and also to avoid potential dead-end approaches and other pitfalls. Ph.D. candidates will likely find that the literature review is one of the most important stages in the dissertation process. To paraphrase Bernard of Chartres and Isaac Newton, you will always reach higher heights when standing on the shoulders of giants.

Step 3: Construct the Hypothesis

The third step is to predict a solution to the problem. In scientific terminology, this is known as specifying the hypothesis. Our market impact model hypothesis includes formulating a mathematical model and incorporating those factors found to influence market price movement. It is important we develop a model that is easily measured and focused. Models that cannot be measured or fail to answer our questions are cast aside.

Step 4: Test the Hypothesis

Step four of the scientific method involves fair, objective, unbiased experiments. Here we perform hypothesis tests on parameters to ensure they are statistically significant and test the overall accuracy of the solution. In